



COMPUTER SCIENCE SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 12)

Time allowed: 20 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Version No.				
1	0	0	7	1

①	●	●	①	①
●	①	①	①	●
②	②	②	②	②
③	③	③	③	③
④	④	④	④	④
⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	●	⑦
⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨

ROLL NUMBER						

①	①	①	①	①	①	①
①	①	①	①	①	①	①
②	②	②	②	②	②	②
③	③	③	③	③	③	③
④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which document outlines a business idea, its goals, and planning details?	Resume	Business plan	Invoice	Advertisement	☐	☐	☐	☐
2. Why is computational thinking important in computer science?	It increases typing speed	It helps break problems into manageable steps	It improves computer hardware	It replaces programming	☐	☐	☐	☐
3. Why is HTML used when creating a website?	To style web pages	To store data	To display text and images	To run programs	☐	☐	☐	☐
4. Which part of development handles the user interface of a website?	Back end	Server side	Database layer	Front end	☐	☐	☐	☐
5. Why are variables used in JavaScript?	To store values that can change	To design web pages	To create hyperlinks	To format text	☐	☐	☐	☐
6. Why is the ALU an important part of the CPU?	It performs arithmetic and logical operations	It manages memory	It controls input devices	It performs analogue linear operations	☐	☐	☐	☐
7. Why is cloud computing needed in modern computing?	To reduce screen size	To replace computers	To access data and services over the Internet	To increase keyboard speed	☐	☐	☐	☐
8. What is meant by fake news?	Verified information	Educational content	False or misleading information	Official announcements	☐	☐	☐	☐
9. What do Sustainable Development Goals (SDGs) represent?	Business profits	Personal goals	Marketing targets	Global development goals	☐	☐	☐	☐
10. How does logical thinking support problem solving?	By using trial and error and providing false results	By arranging steps in a correct sequence	By memorizing commands and repeating these	By designing interfaces	☐	☐	☐	☐
11. Why is software piracy considered illegal?	It violates ownership rights	It is outdated	It slows the working of computer	It limits access to the online data	☐	☐	☐	☐
12. Why is continuous data suitable for measurements like 'heights' of students?	It uses labels	It has fixed values	It is non-numeric	It can take any value in a range	☐	☐	☐	☐

1SA-I 26007-(B) —



COMPUTER SCIENCE SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

Time allowed: 2:40 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

Q. 2 Attempt the following parts.

(10 x 3 = 30)

(i)	What is the main function of the Arithmetic Logic Unit (ALU)? Explain briefly.	03	OR	What is meant by problem decomposition? Elaborate briefly.	03
(ii)	Why is computational thinking important in computer science? Write briefly.	03	OR	How CSS improves the appearance of a web page? Give an example.	2+1
(iii)	State two functions of an operating system.	1.5 + 1.5	OR	What is meant by phishing? Give one example.	2+1
(iv)	Write down two advantages of Big data.	1.5 + 1.5	OR	A computer is running slow while many programs are open. Which hardware component is mainly responsible for processing instruction? Also give reason.	1+2
(v)	Write one harmful effect of using Computing Innovation.	03	OR	How Data Science differs from Data Analytics?	1.5 + 1.5
(vi)	Correct any three errors in the given code of HTML. </html> </head><title>Main page<title> <head> </body> </u>Introduction to Computer </u> <body> <html>	1x3	OR	How packet switching transmits data over a computer network?	03
(vii)	Differentiate between dataset and database.	03	OR	Write down HTML code to apply headings from h1 to h6.	03
(viii)	Why are variables used in Javascript programs? Explain briefly. Give one example.	2+1	OR	Compare natural and artificial systems.	1.5 + 1.5
(ix)	Why identifying inputs, processes, and outputs is necessary?	1x3	OR	Write any three disadvantages of using Artificial Intelligence algorithm.	1x3
(x)	What is meant by design thinking? Also write down the names of five phases.	1.5 + 1.5	OR	Why is cloud computing needed for modern applications? Explain briefly.	03

SECTION – C (Marks 18)

Note: Attempt the following questions.

(3 x 6 = 18)

Q.3	Write down the difference between Half duplex mode and Full duplex mode with one example each. Draw diagrams.	3+3	OR	Briefly discuss the role of AI in 'speech recognition' and 'computer vision'.	3+3
Q.4	Draw a flowchart to find the largest of three unequal numbers.	06	OR	Write down a Javascript code to define two variables and calculate their sum. Also display the result.	06
Q.5	Explain any six benefits of computing innovation.	1x6	OR	What is meant by categorical (qualitative) data? Discuss its main types in detail. Illustrate your answer with examples.	2+4



COMPUTER SCIENCE SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 12)

Time allowed: 20 Minutes

Candidate Sign. _____

Supervisor Sign. _____

Answer Sheet No. _____

Version No.				
1	2	0	7	1

0	0	●	0	0
●	1	1	1	●
2	●	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	●	7
8	8	8	8	8
9	9	9	9	9

ROLL NUMBER							

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

Deleting/overwriting is not allowed.
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Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which strategy explains how a product will reach customers?	Go-to-market strategy	Budget plan	Risk analysis	Product testing	☐	☐	☐	☐
2. Why is algorithmic thinking useful while solving a problem?	It removes the need for inputs	It provides a clear step-by-step solution	It focuses only on outputs	It avoids decision making	☐	☐	☐	☐
3. What does the DOM tree represent in an HTML document?	Program logic, steps and results	Visual design only	Structure of web page elements	Server configuration strategy	☐	☐	☐	☐
4. Which language is most suitable for creating static web pages?	HTML	PHP	SQL	Python	☐	☐	☐	☐
5. Which aspect of CSS can be changed using JavaScript?	Page content	HTML structure	Database records	Font size and color	☐	☐	☐	☐
6. Why is the CPU called the brain of the computer?	It stores data permanently	It displays output	It controls and processes instructions	It supplies power	☐	☐	☐	☐
7. Why is speech recognition considered an important area of AI?	It designs websites	It increases storage capacity	It stores audio files	It allows computers to understand spoken language	☐	☐	☐	☐
8. Which term refers to deceptive emails or messages used to steal personal information?	Hacking	Phishing	Shareware	Freeware	☐	☐	☐	☐
9. What is an entrepreneurial solution mainly designed to do?	Address a problem with a business idea	Increase exams	Reduce technology use and increase manual labor	Copy existing products	☐	☐	☐	☐
10. Why must inputs, processes, and outputs be identified in a problem?	To increase speed	To understand how a solution works	To reduce storage	To design graphics	☐	☐	☐	☐
11. Why are computer ethics important?	To increase speed	To store data	To design software	To guide responsible use	☐	☐	☐	☐
12. Why is categorical data different from numerical data?	It is measured in units	It always changes	It represents categories or labels	It is continuous	☐	☐	☐	☐

—1SA-I 26007-(D)—



COMPUTER SCIENCE SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

Time allowed: 2:40 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

(10 x 3 =30)

Q. 2 Attempt the following parts.

(i)	What is the role of the operating system in a computer system? Elaborate briefly.	03	OR	What is the importance of flowchart in solving a problem?	2+1
(ii)	How logical thinking helps in solving a computational problem? Explain briefly.	03	OR	Differentiate between a static and a dynamic website.	1.5 +1.5
(iii)	Highlight the significance of application software. Give one example.	2+1	OR	What is meant by spyware? Give one example.	2+1
(iv)	Briefly describe quantitative data. Give one example.	2+1	OR	A student performs calculations and logical comparisons in a program. Which part of the CPU carries out these operations? Explain briefly.	1+2
(v)	A news story spreads rapidly online without a reliable source. How can one identify whether it is fake news or credible?	03	OR	Compare Primary and Secondary data collection methods.	1.5 + 1.5
(vi)	Correct any three errors in the given HTML code: </html><head> </title>Heading</title> <head> <body> </h>Welcome to My Website</h> <body> <html>	03	OR	Briefly elaborate two features of star topology. Also draw diagram.	2+1
(vii)	How Data Science differs from Data Analytics?	1.5 + 1.5	OR	Write down HTML code to display ordered list as follows: <u>Memory</u> 1. RAM 2. ROM	1+1 +1
(viii)	Briefly explain the importance of data types in JavaScript. Give one example.	2+1	OR	Why is the Internet called a network of networks? Explain briefly.	03
(ix)	Why is computational thinking important in computer science?	03	OR	Briefly explain one ethical issue related to the use of AI in society.	03
(x)	List and briefly explain any three stages of the design thinking process.	1x3	OR	What is meant by machine learning? Identify its two applications.	2+1

SECTION – C (Marks 18)

Note: Attempt the following questions.

(3x6=18)

Q.3	Compare Asynchronous Transmission and Synchronous Transmission modes. Draw diagrams.	3+3	OR	How is AI used in 'health care' and 'education'? Explain in detail.	3+3
Q.4	Draw a flowchart to find sum and product of five numbers.	06	OR	Write down a JavaScript code to display numbers from 0 to 9.	06
Q.5	Explain the importance of following in safe and responsible use of computers: a. Computer system updated b. Using a strong and unique passwords c. Backing up data	2x3	OR	What is Big data? Explain any two applications of Big data in detail.	2+4



COMPUTER SCIENCE SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 12)

Time allowed: 20 Minutes

Candidate Sign. _____

Supervisor Sign. _____

Answer Sheet No. _____

Version No.				
1	9	0	7	1

0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
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6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

ROLL NUMBER						

0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which stage of design thinking focuses on understanding users' needs?	Ideating	Empathizing	Prototyping	Testing	☐	☐	☐	☐
2. Why is algorithmic thinking useful while solving a problem?	It provides a step-by-step solution	It removes decision making	It focuses only on outputs	It avoids planning	☐	☐	☐	☐
3. Why should users protect personal data online?	To prevent misuse	To share faster	To reduce cost	To improve graphics	☐	☐	☐	☐
4. Which of the following is an example of secondary data collection methods?	Experiment	Observations	Interviews	Government record	☐	☐	☐	☐
5. How does a flowchart help in solving a problem?	It stores data	It executes the program	It visually shows the sequence of steps	It replaces algorithms and performs extra calculations	☐	☐	☐	☐
6. Why is identifying inputs, processes, and outputs important in a problem?	To reduce memory usage	To understand how a solution works	To speed up execution	To create diagrams for visualization of the program	☐	☐	☐	☐
7. Why are hyperlinks important in HTML documents?	They connect one page to another	They change page style	They store images	They run scripts	☐	☐	☐	☐
8. Web development is mainly concerned with:	Computer hardware	Managing networks	Installing software	Creating websites and web applications	☐	☐	☐	☐
9. Why are arrays useful in JavaScript?	To store one value only	To store multiple values in a single variable	To change page style and color	To link web pages	☐	☐	☐	☐
10. Why is a computer considered an artificial system?	It grows naturally	It exists in nature	It is man-made to perform tasks	It works without instructions	☐	☐	☐	☐
11. Why is Artificial Intelligence (AI) useful in solving real-world problems?	It stores large files	It improves hardware speed	It replaces the Internet	It enables machines to make intelligent decisions	☐	☐	☐	☐
12. Which issue arises due to the misuse of personal data online?	Increased speed	Improved usability	Privacy violation	Better communication	☐	☐	☐	☐

—ISA-I 26007-(M)—



COMPUTER SCIENCE SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

56

Time allowed: 2:40 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

(10 x 3 =30)

Q. 2 Attempt the following parts.

(i)	Write down the output of the following Javascript code: <script type= "text / javascript"> var a, b, reward; reward = 5555; a= reward %10; b= reward / 10; document.write (a, "....", b); </script>	03	OR	Why is packet switching efficient for data transmission? Explain briefly.	03
(ii)	How logical thinking helps in solving a computing problem? Explain briefly.	03	OR	What is cloud computing? Why is it needed in modern computing environment?	2+1
(iii)	Enlist any six Sustainable Development Goals (SDGs).	03	OR	How does machine learning help computers solve problems? Give examples of its applications.	2+1
(iv)	What is meant by output device? Write names of any their output devices.	1.5 + 1.5	OR	What is meant by software piracy? Give one example.	2+1
(v)	What is concept map? How does it represent knowledge?	2+1	OR	Give one use of hyperlinks in HTML documents. Also write down the syntax for defining a link.	1+2
(vi)	How is data transferred in synchronous transmission? Also draw diagram.	2+1	OR	Write down the purpose of any two flowchart symbols.	1.5 + 1.5
(vii)	Briefly describe Volume, Velocity and Variety in Big data.	1x3	OR	How Software engineering differs from Hardware engineering?	03
(viii)	A user receives an email asking for bank details with an urgent warning. Identify the type of cyber threat involved. Justify your answer.	1+2	OR	How is data science helpful in solving business problem?	03
(ix)	Briefly explain the role of front-end development in a web project.	03	OR	Why is the Internet considered a global network of networks?	03
(x)	Briefly discuss primary data collection methods. Also identify any two examples.	2+1	OR	Write HTML code to display unordered list as follows: Software 1. System Software 2. Application Software	03

SECTION – C (Marks 18)

Note: Attempt the following questions.

(3x6=18)

Q.3	Explain the importance of following in safe and responsible use of computers: a. Backing up Data b. Use of secure Wi-Fi c. Privacy settings	2x3	OR	What is Big data? Write any four benefits of Big data.	2+4
Q.4	How a star topology differs from a bus topology? Explain. Also draw the diagrams.	3+3	OR	How is Artificial Intelligence (AI) used in 'education' and 'gaming' fields?	3+3
Q.5	For an online order system, identify the inputs, processes, and outputs involved.	2+2 +2	OR	Write down the Javascript code to display numbers from 10 to 1.	06